

SECOND SEMESTER B.A./B.Sc. DEGREE EXAMINATION, APRIL 2020

(CBCSS—UG)

Mathematics

MEC 2C 02—MATHEMATICAL ECONOMICS

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer any number of questions.**Maximum marks : 20.*

1. Define Gini co-efficient.
2. What is meant by Price discrimination ?
3. What is a Non-negativity constraints ?
4. Explain the term Regression.
5. What is an Exogenous variable ?
6. Explain the Open input-output model.
7. State the Young's theorem.
8. What is meant by Local maxima and minima ?
9. Define a Symmetric matrix.
10. What is meant by Continuously differentiable functions ?
11. What is meant by Constrained optimization ?
12. Define Lorenz curve.

Section B

*Answer any number of questions.**Maximum marks : 30.*

13. From the data points, find the equation of the line which best fits the data points (1, 2), (3, 4), (5, 3), (6, 6).
14. Explain the sufficient and necessary conditions for unconstrained optimization.
15. Show whether the following function $x^4 + x^2 + 6xy + 3y^2$ has global minima or maxima.

Turn over

16. Examine whether the input-output system with the following co-efficient matrix is feasible :

$$\begin{bmatrix} 1/2 & 3/5 \\ 1/3 & 5/7 \end{bmatrix}.$$

17. Explain the optimization of a function with several equality constraints.
18. Compute the Hessian matrix of the function $4x^2y - 3xy^3 + 6x$.
19. Describe the Kuhn-Tucker formulation for a constrained minimization problem.

Section C

Answer any one (10 marks)

20. Explain the determination of equilibrium prices in an economy with two sectors using input-output model.
21. Explain the method of least squares and derive the normal equations.